

Grace & Peace Operations Suite

Comprehensive Architecture, Workflows & Operational Specification

A unified, modern enterprise platform built specifically for independent funeral homes, mortuaries, and care directors to orchestrate intakes, tribute generation, family messaging, compliance, and 3D vigils.

1. SYSTEM OVERVIEW & CORE CAPABILITIES

- First-Call Intake & Case Ledger**
Step-by-step intake wizard capturing vital statistics (DOB, DOD, place of death, next-of-kin, service preferences) with real-time case filtering and status tracking.
- Generative AI Obituary Studio**
Multi-tone AI tribute authoring (Traditional, Celebratory, Poetic, Religious, Minimalist) with customizable length, instant regeneration, and inline staff review.
- Multi-Channel Family Communications Hub**
Integrated SMS notifications (Twilio) and professional branded emails (Resend) for instant service coordinates, obituary approvals, and family dispatch.
- Legal Compliance & Document Engine**
Automated state-compliant PDF generator for Cremation/Burial Authorizations, Vital Statistics Worksheets, and Chapel Bulletins using pdf-lib.
- 3D WebGL Perpetual Flame & Memorial Studio**
Interactive Three.js 3D perpetual candlelight, floating remembrance stars, customizable chapel lighting ambiances, and fluid pointer orbit physics.
- Multi-Tenant Row-Level Security (RLS)**
Complete data isolation across funeral homes via Supabase PostgreSQL policies, guaranteeing zero cross-organization record leakage.

2. TECHNOLOGY STACK & ARCHITECTURE

Frontend Application	Next.js 14 (App Router, Server Components & React 18)
Styling & Design System	Tailwind CSS (Bespoke Warm Editorial & Brass Aesthetic)
3D Graphics & Rendering	Three.js (Custom GLSL Shaders, PBR Materials, Lighting)
Database & Auth Engine	Supabase (PostgreSQL with Row Level Security & Auth)
AI Generation Service	OpenAI / OpenRouter API (GPT-4o / Claude 3.5 Sonnet)
PDF Document Engine	PDF-Lib (Real-time dynamic legal document generation)
Telecommunications	Twilio (SMS / Alerts) & Resend (Transactional Email)

3. END-TO-END DIRECTOR WORKFLOW

Step 1: First-Call Intake

When a family calls to report a passing, the director opens the New Case Intake wizard The (/dashboard/cases/new). system captures decedent information, next-of-kin contacts, service type, and SMS opt-in status. A unique case ledger file is automati

Step 2: AI Obituary Drafting & Fine-Tuning

Within the Obituary Studio (/dashboard/cases/[id]/obituary), the director selects tone and presets (Traditional, Poetic, Celebratory) length. The AI crafts a compassionate draft incorporating family memories and vital dates. Staff refine the text with liv

Step 3: Family Review & Multi-Channel Dispatch

Through the Family Communication Hub, the director sends the obituary draft to the family via Resend email and service confirmation alerts via Twilio SMS. Every interaction is logged in the communication ledger with delivery status receipts.

Step 4: Compliance Paperwork & PDF Generation

In the Compliance & Documents view (/dashboard/cases/[id]/documents), the system compiles legal paperwork: Form 104-A (Cremation/Burial Authorization), Vital Statistics Worksheet, and 4-panel Order of Service bulletins ready for physical printing.

Step 5: 3D Perpetual Flame & Family Keepsake

The decedent's digital file features the 3D WebGL Tribute Studio. Family and remote can a relatives light virtual memorial vigil candle, watch remembrance stars ascend, and choose chapel lighting themes in perpetual celebration of their loved one.

4. DATABASE SCHEMA & SECURITY ARCHITECTURE

- **funeral_homes** Tenant entity (name, state, license, contact phone/email).
- **staff_profiles** Staff membership linking auth.users to a specific funeral home ID with director/staff roles.
- **cases** Primary case records: decedent vital stats, informant details, service coordinates, status enum.
- **documents** Obituary revisions, generated PDF storage URLs, compliance forms, and approval signatures.
- **communication_logs** Immutable log of every SMS (Twilio) and Email (Resend) dispatched with delivery receipts.

5. SECURITY & MULTI-TENANT COMPLIANCE

All SQL tables are protected with strict PostgreSQL Row Level Security (RLS). Every query automatically scopes data to the authenticated staff member's funeral_home_id. This guarantees complete data isolation and prevents unauthorized access across different funeral homes or third-party organizations.